1.4.1 SIGNAL DISTORTION IN TRANSMISSION

- Distortion is any change in the shape of a signal's waveform.
- Transmission systems always produce some amount of signal distortion. For the purpose of studying distortion effects on various signals, we'll define two major types of distortion:
  1. Linear distortion
  2. Nonlinear distortion

1. Linear Distortion

- Linear distortion includes any amplitude or delay distortion associated with a linear transmission system.

1.1 **Amplitude distortion:** is easily described in the frequency domain; it means simply that the output frequency components are not in correct proportion.

- Since this is caused by *H(f)* not being constant with frequency.
- Amplitude distortion is sometimes called **frequency distortion.**
- The most common forms of amplitude distortion are excess attenuation or enhancement of extreme high or low frequencies in the signal spectrum.
- If the low-frequency or high-frequency component is attenuated by one-half, the resulting outputs are as shown in Fig. 3.2-4. As expected, loss of the high-frequency term reduces the "sharpness" of the waveform.

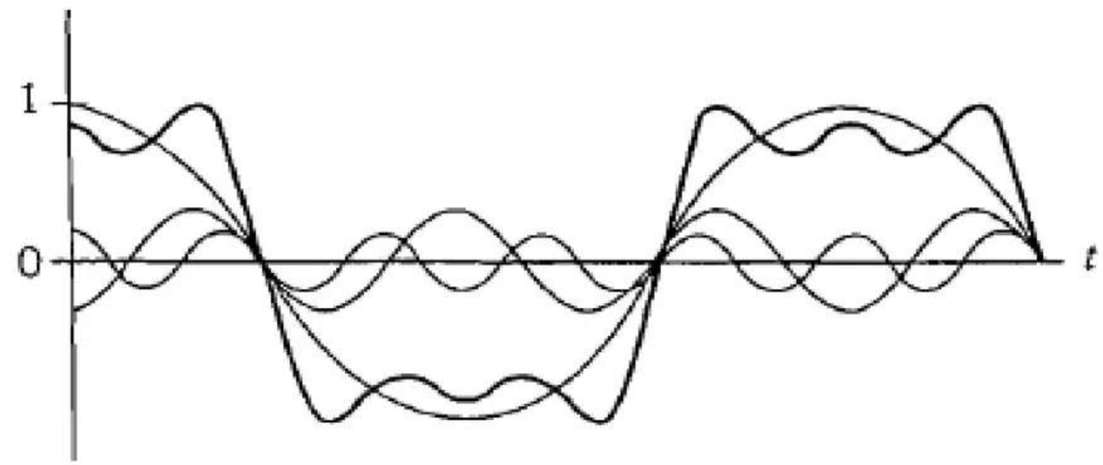

Figure 3.2–3 Test signal $x(t) = \cos \omega_0 t - 1/3 \cos 3\omega_0 t + 1/5 \cos 5\omega_0 t$.

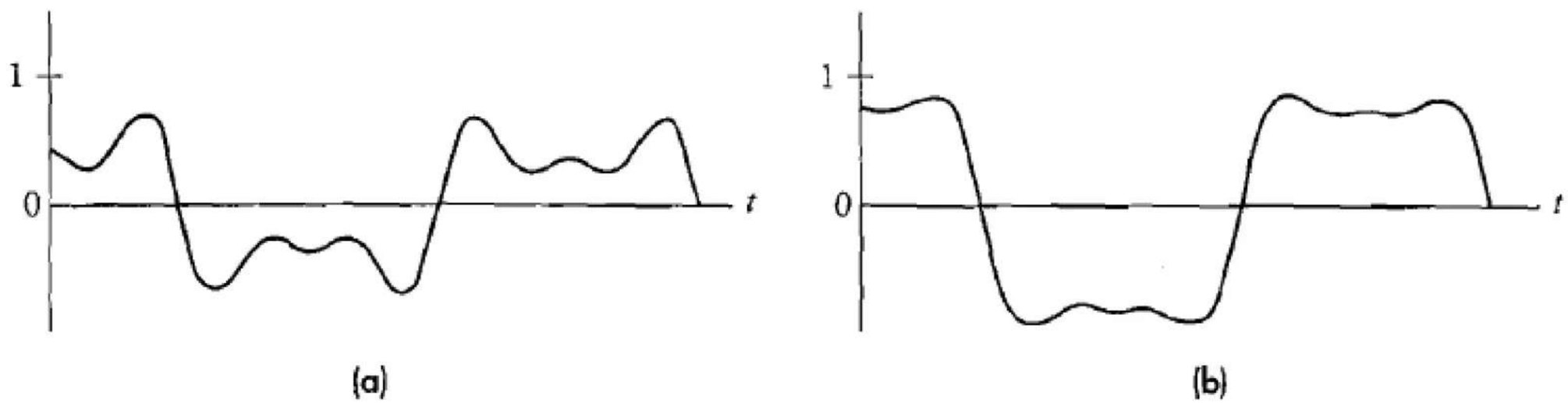

Figure 3.2–4 Test signal with amplitude distortion. *(a)* Low frequency attenuated; *(b)* high frequency attenuated.

1.2 **Phase or Delay distortion**

- The phase shift is not linear, the various frequency components suffer different amounts of time delay, and the resulting distortion is termed phase or delay distortion.
- For an arbitrary phase shift, the time delay is a function of frequency and can be found by writing $\tilde{H}(f) = -2\pi f t_d(f)$ with all angles expressed in *radians*. Thus

$$t_d(f) = -\frac{\arg H(f)}{2\pi f}$$

which is independent of frequency only if arg *H(f)* is linear with frequency.

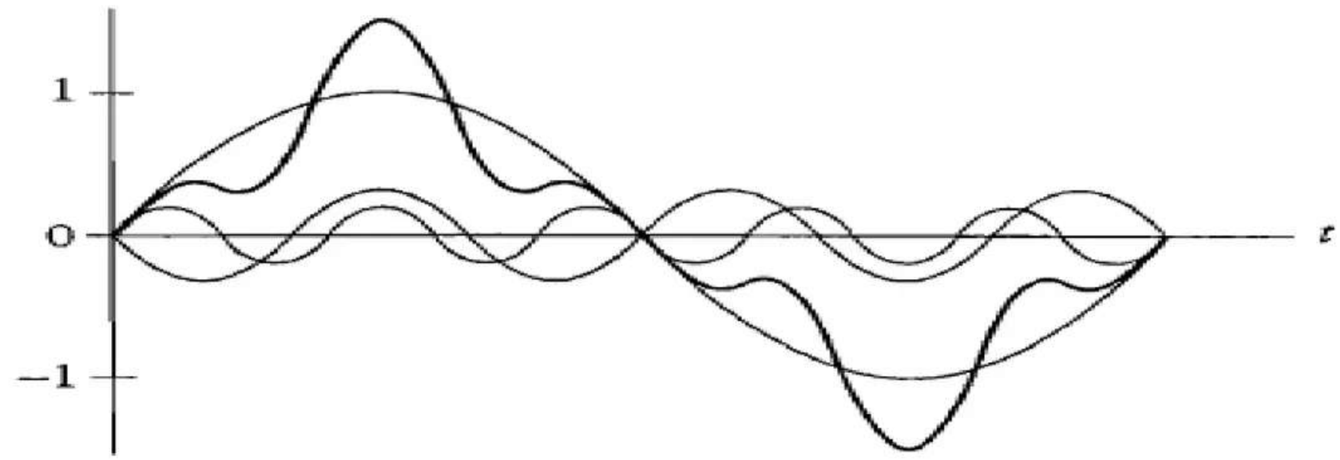

Figure 3.2–5 Test signal with constant phase shift $\theta = -90°$.

Let's take a closer look at the impact of phase delay on a modulated signal. The transfer function of an arbitrary channel can be expressed as

$$H(f) = Ae^{j(-2\pi f t_g + \phi_0)} = (Ae^{j\phi_0})e^{-j2\pi f t_g} \quad [4]$$

where $\arg H(f) = -2\pi f t_g + \phi_0$ leads to $t_d(f) = t_g - \phi_0/2\pi f$ from Eq. (3). If the input to this bandpass channel is

$$x(t) = x_1(t) \cos \omega_c t - x_2(t) \sin \omega_c t \quad [5]$$

then by the time delay property of Fourier transforms, the output will be delayed by t_g. Since $e^{j\phi_0}$ can be incorporated into the sine and cosine terms, the output of the channel is

$$y(t) = Ax_1(t - t_g) \cos [\omega_c(t - t_g) + \phi_0] - Ax_2(t - t_g) \sin [\omega_c(t - t_g) + \phi_0]$$

We observe that $\arg H(f_c) = -\omega_c t_g + \phi_0 = -\omega_c t_d$ so that

$$y(t) = Ax_1(t - t_g) \cos [\omega_c(t - t_d)] - Ax_2(t - t_g) \sin [\omega_c(t - t_d)] \quad [6]$$

From Eq. (6) we see that the carrier has been delayed by t_d and the signals that modulate the carrier, x_1 and x_2, are delayed by t_g. The time delay t_d corresponding to the phase shift in the carrier is called the **phase delay** of the channel. This delay is also sometimes referred to as the **carrier** delay. The delay between the envelope of the

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- In order to recover the original signals Xl and *X2,* the group delay must be constant. Therefore, from Eq. (4) this implies that *tg* can be found directly from the derivative of arg *H(f)* = Θ*(f)* as

$$t_g = -\frac{1}{2\pi}\frac{d\theta(f)}{df}$$

- Note that this condition on arg *H(f)* is less restrictive than in the general case presented earlier. If $\phi_0 = 0$ then the general conditions of distortionless transmission are met and *td* = *tg*

Equalization

- Linear distortion-both amplitude and delay is theoretically curable through the use of *equalization* networks.

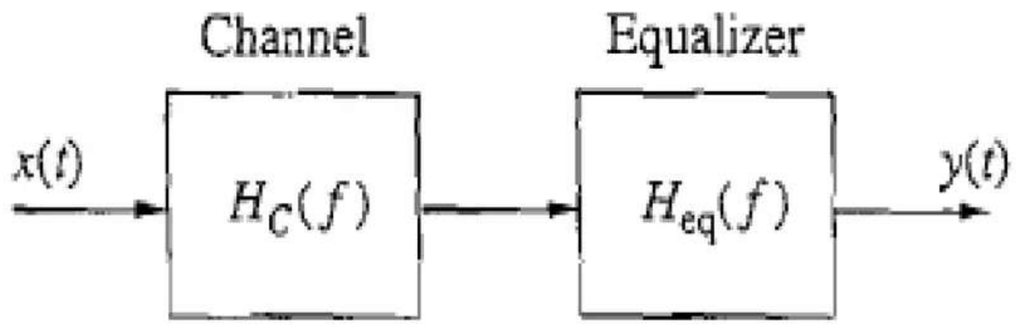

Figure 3.2–6 Channel with equalizer for linear distortion.

- Figure 3.2-6 shows an equalizer *Heq(f)* in cascade with adistorting transmission channel *Hc(f)*.
- Since the overall transfer function is *H(f) = Hc(f)Heq(f)* the final output will be distortionless
  - if *Hc(J)Heq(f)* = $Ke^{-j\omega t_d}$ where *K* and *td* are more or less arbitrary constants. Therefore, we require that

$$H_{eq}(f) = \frac{Ke^{-j\omega t_d}}{H_C(f)}$$

- The tapped-delay-line equalizer, or transversal filter, has emerged as a convenient and flexible device.

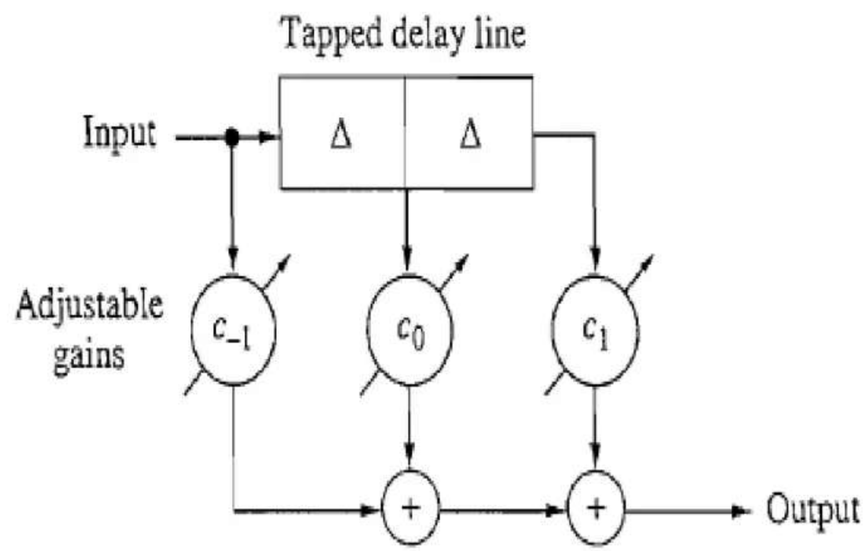

Figure 3.2-8 Transversal filter with three taps.

- To illustrate the principle, Fig. 3.2-8 shows a delay line with total time delay 2 having taps at each end and the middle. The tap outputs are passed through adjustable gains, C_{-1}, Co and C_1 summed to form the final output. Thus

$$y(t) = c_{-1}x(t) + c_0\,x(t-\Delta) + c_1 x(t-2\Delta) \qquad [9a]$$

and

$$\begin{aligned} H_{eq}(f) &= c_{-1} + c_0\,e^{-j\omega\Delta} + c_1 e^{-j\omega 2\Delta} \\ &= \left(c_{-1}e^{+j\omega\Delta} + c_0 + c_1 e^{-j\omega\Delta}\right)e^{-j\omega\Delta} \end{aligned} \qquad [9b]$$

Generalizing Eq. (9*b*) to the case of a $2M\Delta$ delay line with $2M + 1$ taps yields

$$H_{eq}(f) = \left(\sum_{m=-M}^{M} c_m e^{-j\omega m\Delta}\right)e^{-j\omega M\Delta} \qquad [10]$$

Example

Radio systems suffer from multipath distortion caused by two (or more) propagation paths between transmitter and receiver. Reflections due to mismatched impedance on a cable system produce the same effect. suppose the channel output is

$$y(t) = K_1 x(t - t_1) + K_2 x(t - t_2)$$

whose second term corresponds to an *echo* of the first if $t_2 > t_1$. Then

$$H_C(f) = K_1 e^{-j\omega t_1} + K_2 e^{-j\omega t_2} \qquad [11]$$
$$= K_1 e^{-j\omega t_1}(1 + k e^{-j\omega t_0})$$

where $k = K_2/K_1$ and $t_0 = t_2 - t_1$.

If we take $K = K_1$ and $t_d = t_1$ for simplicity in Eq. (8), the required equalizer characteristic becomes

$$H_{eq}(f) = \frac{1}{1 + k e^{-j\omega t_0}} = 1 - k e^{-j\omega t_0} + k^2 e^{-j2\omega t_0} + \cdots$$

The binomial expansion has been used here because, in this case, it leads to the form of Eq. (10) without any Fourier-series calculations. Assuming a small echo, so that $k^2 \ll 1$, we drop the higher-power terms and rewrite $H_{eq}(f)$ as

$$H_{eq}(f) \approx (e^{+j\omega t_0} - k + k^2 e^{-j\omega t_0}) e^{-j\omega t_0}$$

Comparison with Eqs. (9*b*) or (10) now reveals that a three-tap transversal filter will do the job if $c_{-1} = 1$, $c_0 = -k$, $c_1 = k^2$, and $\Delta = t_0$.

2. Nonlinear Distortion and Companding

- sine wave components of the signal are bent (sharply curved or an angle) that the distortion is called nonlinear distortion.
- The instantaneous values of input and output are related by a curve or function *y(t) = T[x(t)],* commonly called the **transfer characteristic.**

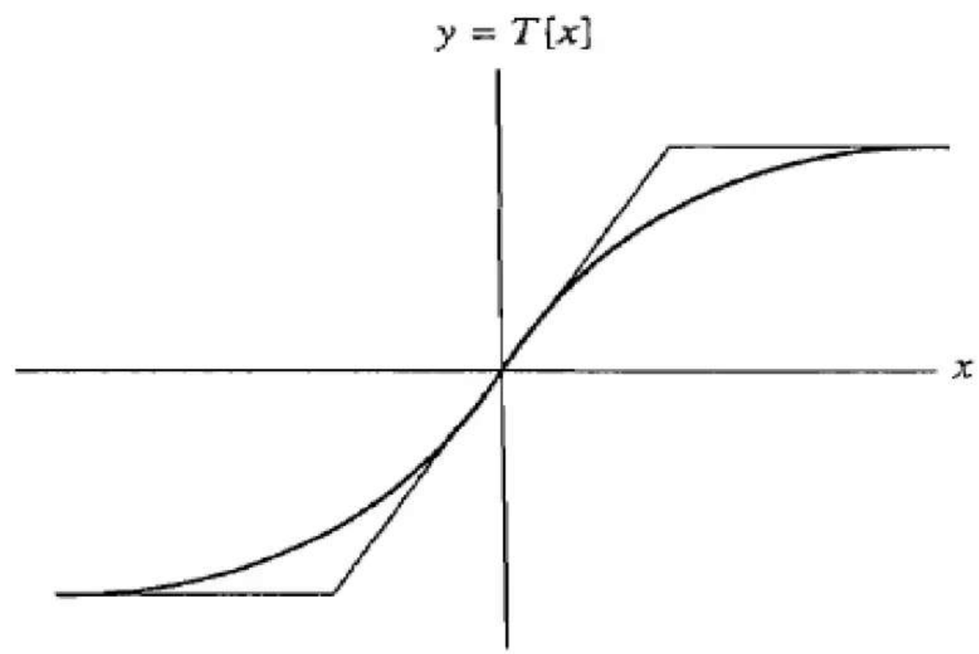

Figure 3.2–9 Transfer characteristic of a nonlinear device.

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots \qquad [12a]$$

and the higher powers of $x(t)$ in this equation give rise to the nonlinear distortion.

Even though we have no transfer function, the output spectrum can be found, at least in a formal way, by transforming Eq. (12a). Specifically, invoking the convolution theorem,

$$Y(f) = a_1 X(f) + a_2 X * X(f) + a_3 X * X * X(f) + \cdots \qquad [12b]$$

Now if $x(t)$ is bandlimited in W, the output of a linear network will contain no frequencies beyond $|f| < W$. But in the nonlinear case, we see that the output includes $X * X(f)$, which is bandlimited in $2W$, $X * X * X(f)$, which is bandlimited in $3W$, and so on. The nonlinearities have therefore created output frequency components that were not present in the input. Furthermore, since $X * X(f)$ may contain components for $|f| < W$, this portion of the spectrum overlaps that of $X(f)$. Using filtering techniques, the added components at $|f| > W$ can be removed, but there is no convenient way to get rid of the added components at $|f| < W$. These, in fact, constitute the nonlinear distortion.

Companding

- The joint use of compressing and expanding is called companding.
- High amplitude analog signals are compressed prior to transmission and then expanded in the receiver.
- A compressor has greater amplification at low signal levels than at high signal levels, similar to Fig. 3.2-9, and thereby compresses the range of the input signal. If the compressed signal falls within the linear range of the channel, the signal at the channel output is proportional to *Tcomp[x(t)]* which is distorted by the compressor but not the channel. Ideally, then, the expander has a characteristic that perfectly complements the compressor so the expanded output is proportional to *Texp{Tcomp[x(t)]}* = *x(t),* as desired.

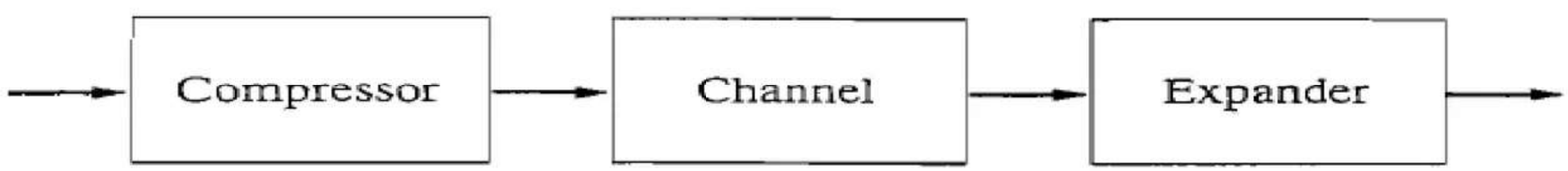

Figure 3.2–10 Companding system.

- Besides reducing nonlinear distortion, companding tends to compensate for the signal-level difference between loud and soft talkers.
- The key advantage of companding compared to the simpler technique of linearly attenuating the signal at the input (to keep it in the linear range of the channel) and linearly amplifying it at the output.